MSG No. 93.120 $P4_2221'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 222.1'

No.	position	mapping
1	$[0, 0, 0]$	[1,4,5,6,9,12,13,14]
2	$\left[0, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 222.1'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]

Table 3: Wyckoff site: 2c, site symmetry: 222.1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]

Table 4: Wyckoff site: 2d, site symmetry: 222.1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,4,5,6,9,12,13,14]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,3,7,8,10,11,15,16]

Table 5: Wyckoff site: 2e, site symmetry: 2.221'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,6,7,8,9,14,15,16]
2	$\left[0, 0, \frac{3}{4}\right]$	[2,3,4,5,10,11,12,13]

Table 6: Wyckoff site: 2f, site symmetry: 2.221'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[1,6,7,8,9,14,15,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[2,3,4,5,10,11,12,13]

Table 7: Wyckoff site: 4g, site symmetry: 2..1'

No.	position	mapping
1	$[0, 0, z]$	[1,6,9,14]
2	$\left[0, 0, z + \frac{1}{2}\right]$	[2,3,10,11]

continued ...

Table 7

No.	position	mapping
3	$[0,\ 0,\ -z]$	`[4,5,12,13]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,6,9,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,12,13]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8,15,16]`

Table 9: Wyckoff site: `4i`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,9,14]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,10,11]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5,12,13]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8,15,16]`

Table 10: Wyckoff site: `4j`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4,9,12]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,7,10,15]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[3,8,11,16]`
4	$[-x,\ 0,\ 0]$	`[5,6,13,14]`

Table 11: Wyckoff site: `4k`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,4,9,12]`
2	$\left[\frac{1}{2},\ x,\ 0\right]$	`[2,7,10,15]`
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[3,8,11,16]`
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,13,14]`

Table 12: Wyckoff site: `4l`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,4,9,12]`
2	$[0,\ x,\ 0]$	`[2,7,10,15]`
3	$[0,\ -x,\ 0]$	`[3,8,11,16]`

continued ...

Table 12

No.	position	mapping
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[5,6,13,14]

Table 13: Wyckoff site: 4m, site symmetry: .2.1'

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	[1,4,9,12]
2	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[2,7,10,15]
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[3,8,11,16]
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	[5,6,13,14]

Table 14: Wyckoff site: 4n, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	[1,7,9,15]
2	$\left[-x,\, x,\, \frac{3}{4}\right]$	[2,5,10,13]
3	$\left[x,\, -x,\, \frac{3}{4}\right]$	[3,4,11,12]
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[6,8,14,16]

Table 15: Wyckoff site: 4o, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\, x,\, \frac{3}{4}\right]$	[1,7,9,15]
2	$\left[-x,\, x,\, \frac{1}{4}\right]$	[2,5,10,13]
3	$\left[x,\, -x,\, \frac{1}{4}\right]$	[3,4,11,12]
4	$\left[-x,\, -x,\, \frac{3}{4}\right]$	[6,8,14,16]

Table 16: Wyckoff site: 8p, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,9]
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[2,10]
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[3,11]
4	$[x,\, -y,\, -z]$	[4,12]
5	$[-x,\, y,\, -z]$	[5,13]
6	$[-x,\, -y,\, z]$	[6,14]
7	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[7,15]
8	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[8,16]